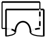
$Z($  $) = \varepsilon\colon A/[A,A] \to \mathbf{k}$

$Z($  $)\colon {}_{A_1}A_{A_2} \otimes {}_{A_3^{\mathrm{op}}}A_{A_4^{\mathrm{op}}} \to {}_{A_1}A_{A_3} \otimes {}_{A_4}A_{A_2} : a \otimes b \mapsto \sum ax_i \otimes bx^i$

$Z($  $)\colon {}_{A_1}A_{A_2} \otimes {}_{A_3^{\mathrm{op}}}A_{A_4^{\mathrm{op}}} \to {}_{A_1}A_{A_3} \otimes {}_{A_4}A_{A_2} : a \otimes b \mapsto \sum ax_i \otimes bx^i$

$Z($  $) = [w^{-1}] \in A/[A,A]$